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<h1 align="center">📹 Smart Attendance System</h1>

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Real-Time Face Recognition · Anti-Spoofing Liveness Detection · AI-Powered Chatbot

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<img src="https://img.shields.io/badge/OpenCV-4.x-5C3EE8?style=for-the-badge&logo=opencv&logoColor=white" alt="OpenCV">

<img src="https://img.shields.io/badge/Streamlit-1.53+-FF4B4B?style=for-the-badge&logo=streamlit&logoColor=white" alt="Streamlit">

<img src="https://img.shields.io/badge/Gemini-2.5_Flash-4285F4?style=for-the-badge&logo=google&logoColor=white" alt="Gemini">

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An end-to-end, AI-powered attendance solution that leverages **OpenCV** for real-time facial recognition and **anti-spoofing liveness detection**. The system automates the check-in process through a sleek Streamlit dashboard and features an integrated Gemini-powered chatbot for natural language record querying.

✨ Features

| Feature | Description |

|---|---|

| 🔍 **Advanced Face Recognition** | High-accuracy matching using **LBPH** and **DNN-based embeddings** |

| 🛡️ **Anti-Spoofing Liveness** | Blink-frequency analysis to ensure a live person, not a photo |

| 🤖 **AI Attendance Assistant** | **RAG chatbot** powered by Gemini 2.5 Flash + FAISS for natural language queries |

| 📊 **Automated Analytics** | Daily/monthly reports with visualizations, exportable to **PDF, Excel, and CSV** |

| ⚙️ **Live Enrollment** | Register students directly via camera with automated dataset training |

🛠️ Tech Stack

| Category | Tools |

|---|---|

| **Computer Vision** | OpenCV, DeepFace, Imutils |

| **Frontend** | Streamlit, Plotly |

| **LLM** | Google Gemini 2.5 Flash |

| **Database & Vector Search** | SQLite3, FAISS |

| **Data Processing** | Pandas, NumPy, Scikit-Learn |

📂 Project Structure

```
📦 Smart-Attendance-System
│
├── main.py                  # Entry point — initializes dirs & launches app
├── dashboard.py             # Streamlit UI (live feed, reports, controls)
├── face_recognition_system.py   # Core OpenCV + liveness detection logic
├── chatbot.py               # Gemini + FAISS RAG chatbot
├── database.py              # SQLite handling & report generation
├── embedding_pipeline.py       # Face embeddings + ML (SVM) training
├── requirements.txt          # Python dependencies
│
└── data/                    # Auto-generated at runtime
    ├── attendance.db         # SQLite attendance database
    ├── models/               # Trained LBPH & SVM models
    └── embeddings/           # Facial embedding vectors
```

🚀 Quick Start

Prerequisites

- Python 3.9+
- A webcam
- A [Google Gemini API key](https://aistudio.google.com/app/apikey)

1. Clone the Repository

```bash
git clone https://github.com/Zerodistraction-max/Smart-Attendance-with-Face-Liveness-Detection.git
cd Smart-Attendance-with-Face-Liveness-Detection
```

2. Install Dependencies

```bash
pip install -r requirements.txt
```

3. Environment Setup

Create a `.env` file in the root directory and add your credentials:

```env
GOOGLE_API_KEY=your_gemini_api_key_here
ATTENDANCE_DB_PATH=data/attendance.db
```

4. Run the Application

```bash
python main.py
```

> The system will automatically initialize all required directories and launch the Streamlit dashboard in your browser.

💻 How It Works

```
┌──────────────┐   ┌───────────────────────┐   ┌──────────────────┐
│  Live Camera  │───▶│  Liveness Detection   │───▶│ Face Recognition │
│     Feed      │   │  (Blink Frequency)    │   │  (LBPH / DNN)    │
└──────────────┘   └───────────────────────┘   └────────┬─────────┘
                                                         │
                        ┌────────────────────────────────▼────┐
                        │         SQLite Database             │
                        │  (Attendance Logs + Student Records) │
                        └─────────────────────────────────────┘
                                        │
                     ┌──────────────────┴───────────────────┐
                     │                                      │
                ┌────▼──────────┐               ┌───────────▼─────────┐
                │  Analytics &   │               │  Gemini RAG Chatbot  │
                │  Report Export │               │  (FAISS + Gemini 2.5) │
                │  (PDF/Excel/CSV)│               └─────────────────────┘
                └───────────────┘
```

1. **Live Feed** — The camera captures a real-time video stream.
2. **Liveness Check** — Blink detection confirms the subject is a live person.
3. **Recognition** — LBPH or DNN embeddings identify the student.
4. **Logging** — A timestamped record is saved to SQLite.
5. **Analytics** — Reports are auto-generated and available for export.
6. **Chatbot** — Ask questions in plain English like *"Who was absent on Monday?"*

🤖 AI Chatbot Examples

The Gemini-powered RAG assistant can answer natural language queries over your attendance database:

> 💬 *"Was Rahul present yesterday?"*

> 💬 *"How many students were absent last week?"*

> 💬 *"Show me the attendance summary for March."*

> 💬 *"Who has the lowest attendance this month?"*

📊 Reports & Analytics

- **Daily Summary** — Per-day attendance breakdown
- **Monthly Report** — Aggregated monthly statistics with charts
- **Export Formats** — PDF, Excel (`.xlsx`), and CSV
- **Visualizations** — Bar charts and trend graphs via Plotly

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🔧 Configuration

| Variable | Description | Default |

|---|---|---|

| `GOOGLE_API_KEY` | Your Gemini API key | *(required)* |

| `ATTENDANCE_DB_PATH` | Path to SQLite database | `data/attendance.db` |

🤝 Contributing

Contributions are welcome! Here's how to get started:

1. **Fork** the repository
2. **Create** a feature branch: `git checkout -b feature/your-feature-name`
3. **Commit** your changes: `git commit -m 'Add some feature'`
4. **Push** to the branch: `git push origin feature/your-feature-name`
5. **Open** a Pull Request

Feel free to open an [issue](https://github.com/Zerodistraction-max/Smart-Attendance-with-Face-Liveness-Detection/issues) for bug reports, feature requests, or questions.

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